

Isobaric vapor–liquid equilibrium of trifluoroacetic acid + water, trifluoroacetic acid + ethyl trifluoroacetate and ethyl trifluoroacetate + ethanol binary mixtures

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ABSTRACT

Vapor–liquid equilibrium data for ethyl trifluoroacetate + trifluoroacetic acid, ethyl trifluoroacetate + ethanol, and trifluoroacetic acid + water at the pressure 101.3 kPa were measured with a modified Rose still. The thermodynamic consistency of the VLE data was confirmed by Herrington's method. The VLE data were correlated by the NRTL and Wilson models with satisfactory results, and the interaction parameters of these mixtures were estimated.

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1. Introduction

Ethyl trifluoroacetate (ETFA) contains a trifluoroacetyl group, which can be used as an amines protecting group in organic synthesis [1]. So ETFA has been recognized as an important fine chemical intermediate and widely used for preparing a variety of pesticides, pharmaceuticals, dyes, and liquid crystal. ETFA can also be used as solvent in the manufacture of other fine chemicals [2]. There are several methods which can be used to produce ETFA. ETFA can be synthesized through the substitution reaction of trifluoroacetyl chloride with ethanol (EtOH), where a substantial stoichiometric excess of trifluoroacetyl chloride is used [3]. ETFA can also be prepared through the esterification reaction of trifluoroacetic acid (TFA) with EtOH by using a liquid hydrofluoric acid as a catalyst. Taking account of the toxicity and cost, however, the esterification reactions of TFA with EtOH using homogeneous catalysts has become the main method for producing ETFA nowadays.

Because the esterification reaction is a typical equilibrium-limited reaction, the conversion rate of TFA is strongly limited by thermodynamic equilibrium constrains. So the separation and purification process (distillation was preferred) will be adopted in

order to obtain ETFA with desired purity. As a common separating process, accurate vapor–liquid equilibrium (VLE) data are very important and necessary for the design and operation of the distillation [4–6]. Up to now, however, the VLE data for this system except EtOH–H₂O have never been reported in the literature.

Therefore, in this work isobaric VLE data for three binary mixtures, including ETFA + TFA, ETFA + EtOH, and TFA + H₂O, were measured at 101.33 kPa with a modified Rose still. Then, both NRTL and Wilson models were used to correlate the experimental VLE data.

2. Experimental

2.1. Chemicals

Both trifluoroacetic acid (99.5%, mass%) and ethyl trifluoroacetate (99.0%, mass%) were provided by Aladdin Co., Ltd. Ethanol (99.7%, mass%) was purchased from Sinopharm Chemical Reagent Co., Ltd., China. The purity of the chemicals was verified by gas chromatography and no appreciable peaks of impurities were detected. The water used was purified from an ultrafiltration system in our laboratory. The density, refractive index, and normal boiling point of the chemicals were determined by an density meter (Anton Paar DMA-58), an Abbe refractometer (Atago 3T), and

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Nomenclature		y_i	Vapor mole fraction of component i
A_i, B_i, C_i	Antoine constants of pure component i	Z	Correction factor
F	Objective function	<i>Greek letters</i>	
K	The dimerization equilibrium constant	α	Non-randomness parameter in the NRTL model
N	Number of the data points	ρ	Density, g cm ⁻³
n_D	Refractive index	γ_i	Activity coefficients of component i
P	Pressure, kPa	Δ	Model parameters of Wilson, J mol ⁻¹
P^s	Saturated vapor pressure of pure component, kPa	Δg	Energy parameters in the NRTL model, J mol ⁻¹
R	Gas constant	φ	Vapor-phase fugacity coefficient of component
T	Absolute temperature, K	<i>Subscripts</i>	
T_b	Boiling temperature of pure component at 101.3 kPa, K	i, j	Components i and j
u	Uncertainty	cal	Calculated data
V	Molar volume	exp	Experimental data
x_i	Liquid mole fraction of component i		

the modified Rose still, respectively. The physical properties of the chemicals used in this work are listed in Table 1. It can be found that the measured physical property data of the chemical were basically consistent with literature values. So no further purification was performed for these chemicals.

2.2. Apparatus and procedure

Vapor–liquid equilibrium data were measured by the double circulating vapor–liquid equilibrium still (a modified Rose still) at atmospheric pressure. Details of this apparatus were given in our previous studies [5,22]. In a typical experiment, pure component 1 (light component) was introduced into the equilibrium cell firstly. Equilibrium between vapor and liquid phases was assumed when the equilibrium temperature remained constant for at least 30 min. Then, the readings of pressure and temperature of the equilibrium cell were recorded manually, and the vapor and liquid samples were withdrawn simultaneously for analysis. Next, a pre-determined volume of liquid was withdrawn from the equilibrium cell and an equal volume of pure component 2 (heavy component) was added the cell. The vapor and the liquid phases were continuously circulating to provide intimate contact of the two phases and ensure that equilibrium can be established. The pressure and temperature of the equilibrium cell were read, and the vapor and liquid samples were withdrawn and analyzed during all runs. Repeating the above procedures, the additions of component 2 were continued until the target composition was reached. The concentration of ETFA and EtOH were analyzed by Gas Chromatography with dioxane as internal, while the TFA concentration was checked by titration with 0.1 N sodium hydroxide using phenolphthalein as indicator. Each measurement was repeated 3 times and averaged as the final result.

The equilibrium temperature was determined with a calibrated precision thermometer with maximum error of 0.1 K. The pressure in the apparatus was measured by means of a precision barometer with an uncertainty of 0.1 kPa (Jiangshan Glassware Instrument & Meter Factory, DYM3, China). Because the atmospheric pressure may change a little during the experiments, the equilibrium temperature was corrected to 101.3 kPa according to the method proposed by Hiaki and Kawai [23].

2.3. Reliability test of the experimental apparatus

To check the reliability of the experimental apparatus, the VLE data of methanol + isopropanol binary mixture at 101.3 kPa were determined. The experimental and literature data are presented in Fig. 1. The average relative deviations for equilibrium temperature and vapor molar fraction are 0.02% and 0.88%, respectively. The result shows that the validation data are in good agreement with the literature values [5], which illustrates that the data obtained by this method and the apparatus are repeatable and reliable.

3. Result and discussion

3.1. Vapor liquid equilibrium measurements

The VLE data for TFA–H₂O, ETFA–EtOH, and ETFA–TFA binary systems were measured at 101.3 kPa and presented in Tables 2–4, respectively. The T – x – y diagrams of three binary systems are showed in Figs. 2–4, respectively. Notice that there all have intersecting points between the bubble point curve and the dew point curve over the entire composition range in three T – x – y diagrams, which demonstrates that all three binary systems exhibit deviations from Raoult's law and present azeotropic behaviors.

Table 1
Physical properties of pure compounds.

Compound	T_b (K) (101.3 kPa)		ρ (g/cm ³) (293.15 K)		n_D (293.15 K)	
	This work	Literature	This work	Literature	This work	Literature
ETFA	335.65	335.15 [7] 334.85 [8]	1.1897	1.1900 [7] 1.1913 [9]	1.3086	1.3079 [10] 1.3077 [11]
EtOH	351.55	351.65 [12] 351.45 [13]	0.7891	0.7893 [14] 0.7898 [15]	1.3634	1.3618 [16]
TFA	344.91	344.93 [17]	1.4891	1.4890 [18]	1.2852	1.2850 [19]
H ₂ O	373.15	373.15 [20]	0.9981	0.9982 [21]	1.3332	1.3330 [19]

$u(T_b) = \pm 0.1$ K; $u(\rho) = \pm 0.0001$ g/cm³; $u(n_D) = \pm 0.0001$.

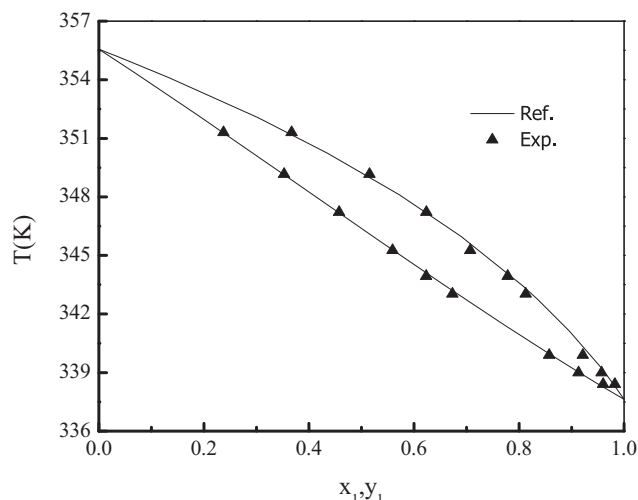


Fig. 1. T–x–y diagram for methanol (1)–isopropanol (2) at 101.3 kPa.

3.2. Vapor–liquid equilibrium model

The general equilibrium relationship for vapor–liquid equilibrium can be expressed as follows:

$$y_i \hat{\phi}_i^v P = x_i \gamma_i \phi_i^s P_i^s \exp \left[\frac{V_i^l (P - P_i^s)}{RT} \right] \quad (1)$$

At low or moderate pressures, $\exp[V_i^l (P - P_i^s)/RT] \approx 1$, and the vapor phase could be treated as ideal gas which resulted in $\hat{\phi}_i^v = 1$

Table 2
Vapor–liquid equilibrium data of TFA(1)–H₂O(2) at 101.3 kPa.

No.	T (K)	x ₁	y ₁	Wilson			NRTL		
				γ ₁	γ ₂	T _{cal} (K)	γ ₁	γ ₂	T _{cal} (K)
1	344.91	1.000	1.000	1.000	0.013	344.95	1.000	0.020	344.95
2	345.35	0.984	0.996	0.997	0.018	345.50	0.998	0.025	345.47
3	347.15	0.947	0.991	0.975	0.037	347.29	0.982	0.042	347.08
4	348.44	0.926	0.987	0.956	0.051	348.49	0.966	0.053	348.17
5	349.66	0.899	0.984	0.927	0.072	350.21	0.942	0.071	349.77
6	350.95	0.870	0.979	0.893	0.099	352.20	0.910	0.094	351.70
7	355.15	0.833	0.990	0.850	0.142	354.93	0.865	0.133	354.44
8	357.75	0.788	0.983	0.795	0.198	358.31	0.806	0.185	357.99
9	359.00	0.776	0.974	0.781	0.214	359.23	0.790	0.201	358.97
10	361.20	0.744	0.964	0.743	0.258	361.69	0.746	0.245	361.65
11	362.43	0.727	0.955	0.723	0.283	363.01	0.723	0.271	363.09
12	363.53	0.719	0.947	0.715	0.294	363.58	0.713	0.283	363.73
13	364.62	0.700	0.940	0.693	0.321	364.96	0.688	0.311	365.24
14	366.63	0.680	0.930	0.672	0.350	366.40	0.663	0.343	366.83
15	368.55	0.655	0.902	0.645	0.387	368.16	0.632	0.383	368.76
16	370.12	0.631	0.865	0.619	0.422	369.82	0.601	0.422	370.55
17	371.52	0.614	0.838	0.602	0.446	370.89	0.582	0.450	371.70
18	372.99	0.593	0.806	0.580	0.476	372.18	0.558	0.484	373.06
19	374.15	0.575	0.789	0.561	0.503	373.27	0.536	0.514	374.17
20	376.23	0.542	0.693	0.528	0.549	374.99	0.501	0.566	375.88
21	377.85	0.519	0.623	0.507	0.580	376.03	0.478	0.601	376.87
22	378.55	0.489	0.569	0.476	0.620	377.25	0.447	0.645	377.97
23	378.85	0.462	0.526	0.448	0.656	378.14	0.420	0.684	378.71
24	378.97	0.416	0.445	0.403	0.713	379.20	0.377	0.744	379.47
25	377.83	0.309	0.221	0.306	0.832	379.63	0.290	0.862	379.29
26	376.65	0.249	0.150	0.257	0.888	378.86	0.249	0.912	378.37
27	375.63	0.201	0.112	0.222	0.925	377.92	0.220	0.944	377.43
28	374.77	0.157	0.082	0.193	0.954	376.89	0.197	0.967	376.47
29	373.83	0.069	0.043	0.142	0.991	374.71	0.158	0.994	374.54
30	373.43	0.020	0.014	0.118	0.999	373.57	0.139	1.000	373.54
31	373.17	0.004	0.002	0.111	1.000	373.23	0.134	1.000	373.22

$u(T) = \pm 0.1$ K, and $u(x_i) = u(y_i) = \pm 0.001$.

Table 3
Vapor–liquid equilibrium data of ETFA(1)–EtOH(2) at 101.3 kPa.

No.	T(K)	x ₁	y ₁	Wilson			NRTL		
				γ ₁	γ ₂	T _{cal} (K)	γ ₁	γ ₂	T _{cal} (K)
1	336.23	1.000	1.000	1.000	12.604	335.99	1.000	8.380	335.76
2	334.77	0.992	0.985	1.000	11.490	334.85	1.000	8.022	335.06
3	333.77	0.969	0.954	1.006	8.992	332.89	1.003	7.031	333.58
4	332.46	0.940	0.906	1.018	6.992	331.68	1.011	6.033	332.40
5	331.47	0.889	0.852	1.052	4.913	330.77	1.035	4.696	331.29
6	331.08	0.859	0.840	1.078	4.156	330.54	1.056	4.107	330.96
7	330.78	0.818	0.798	1.119	3.431	330.37	1.090	3.481	330.71
8	330.56	0.723	0.771	1.238	2.437	330.27	1.201	2.515	330.55
9	330.65	0.642	0.746	1.368	1.965	330.33	1.330	2.017	330.58
11	331.49	0.471	0.710	1.764	1.423	330.79	1.742	1.431	330.89
12	331.93	0.408	0.693	1.969	1.304	331.16	1.957	1.305	331.18
13	332.46	0.342	0.670	2.235	1.208	331.73	2.233	1.204	331.69
14	332.97	0.303	0.663	2.427	1.161	332.23	2.430	1.156	332.15
15	333.74	0.254	0.631	2.708	1.113	333.05	2.714	1.107	332.97
16	334.34	0.222	0.618	2.920	1.087	333.76	2.923	1.081	333.69
17	335.55	0.182	0.575	3.237	1.058	334.96	3.228	1.054	334.93
18	336.35	0.156	0.548	3.466	1.043	335.92	3.442	1.039	335.94
19	336.72	0.142	0.527	3.606	1.036	336.55	3.571	1.033	336.60
20	338.15	0.124	0.493	3.793	1.028	337.48	3.737	1.025	337.57
21	339.27	0.096	0.450	4.121	1.017	339.22	4.024	1.015	339.39
22	340.13	0.087	0.422	4.234	1.014	339.90	4.119	1.012	340.08
23	341.35	0.072	0.387	4.450	1.009	341.24	4.297	1.008	341.46
24	342.36	0.063	0.354	4.569	1.007	342.05	4.391	1.006	342.28
25	343.85	0.041	0.308	4.913	1.003	344.52	4.665	1.003	344.77
26	344.74	0.041	0.274	4.920	1.003	344.63	4.663	1.003	344.88
27	345.93	0.032	0.236	5.058	1.002	345.76	4.764	1.002	345.99
28	346.75	0.026	0.213	5.160	1.001	346.63	4.838	1.001	346.85
29	347.60	0.021	0.176	5.255	1.001	347.49	4.906	1.001	347.69
30	348.53	0.018	0.128	5.305	1.001	348.00	4.936	1.001	348.17
31	349.50	0.009	0.093	5.476	1.000	349.61	5.059	1.000	349.72
32	350.50	0.005	0.050	5.563	1.000	350.53	5.115	1.000	350.59

$u(T) = \pm 0.1$ K, and $u(x_i) = u(y_i) = \pm 0.001$.

and $\phi_i^s = 1$. So, Eq. (1) can be simplified and rewritten as,

$$y_i P = x_i \gamma_i P_i^s \quad (2)$$

where y_i, x_i are the mole fraction of component i in the vapor and liquid phase, respectively; γ_i is the activity coefficients of component i ; P is the total pressure; P_i^s is the saturated vapor pressure of pure component i and can be calculated by Antoine equation. Those constants of the components i are listed in Table 5. The liquid-phase non ideality can be considered by activity coefficient model.

In two binary systems (TFA–H₂O and ETFA–TFA), since TFA may form dimers in the vapor phase by means of hydrogen bonds between two molecules, the vapor phase fugacity coefficients in Eq. (1) could not be estimated considering as ideal behavior. For substances such as carboxylic acids which associate very strongly, the virial equation is not valid. However, the “chemical theory” for nonideality can give good predictions in such case when an equilibrium constant for association is available [25]. According to the Marek’s chemical theory [26], there are monomer and dimer TFA molecules in vapor phases. In order to transfer the problem of chemical equilibrium into the problem of phase equilibrium, a correction factor Z is introduced into the vapor–liquid equilibrium relationship, which is given by

$$y_i Z_i P = x_i \gamma_i P_i^s \quad (3)$$

The correction factor Z can be calculated according to the method of Marek [26], where Z_A is used for the associating component:

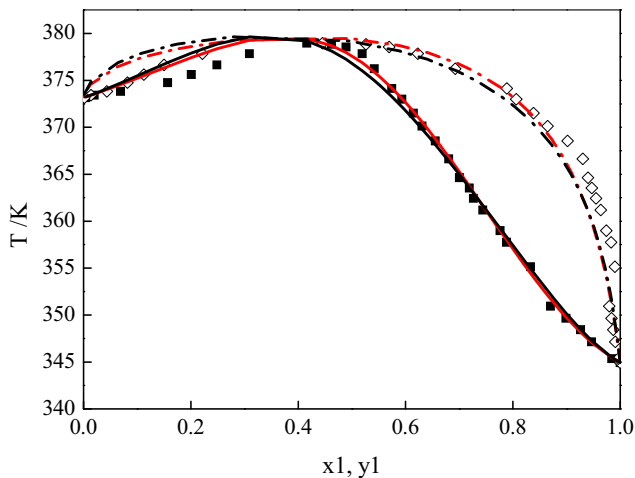
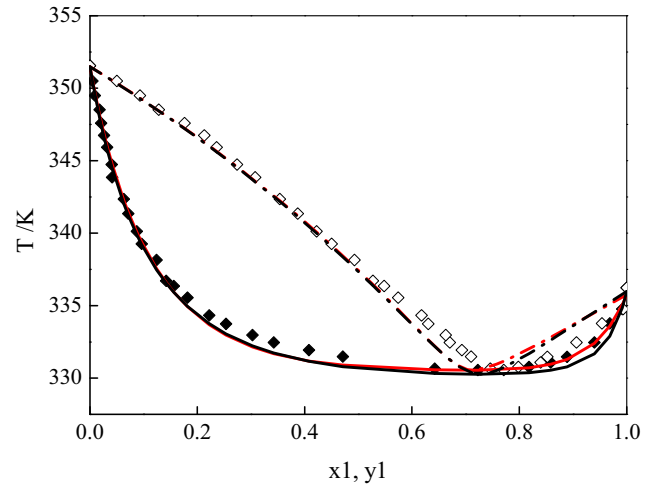
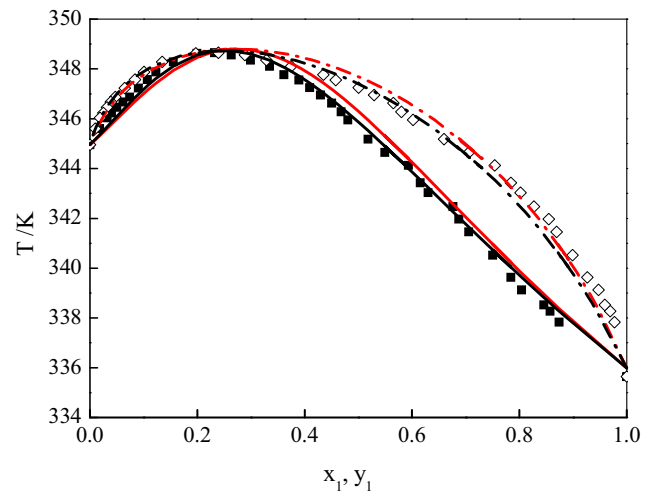
Table 4

Vapor–liquid equilibrium data of ETFA(1)–TFA(2) at 101.3 kPa.

No.	T(K)	x_1	y_1	Wilson			NRTL		
				γ_1	γ_2	$T_{cal}(K)$	γ_1	γ_2	$T_{cal}(K)$
1	335.65	1.000	1.000	1.000	0.527	335.99	1.000	0.488	335.99
2	337.83	0.874	0.977	0.995	0.570	338.27	0.994	0.534	338.37
3	338.27	0.857	0.969	0.993	0.576	338.59	0.992	0.542	338.71
4	338.52	0.845	0.959	0.992	0.580	338.83	0.991	0.547	338.95
5	339.13	0.804	0.947	0.987	0.596	339.63	0.985	0.564	339.81
6	340.52	0.750	0.899	0.977	0.618	340.73	0.973	0.589	340.96
7	339.63	0.784	0.927	0.984	0.604	340.04	0.981	0.573	340.23
8	341.46	0.706	0.869	0.967	0.637	341.65	0.961	0.611	341.93
9	341.97	0.687	0.855	0.962	0.645	342.04	0.955	0.621	342.34
10	342.47	0.676	0.827	0.959	0.650	342.26	0.951	0.627	342.57
11	343.03	0.630	0.802	0.943	0.671	343.24	0.933	0.651	343.59
12	343.43	0.615	0.784	0.938	0.678	343.55	0.926	0.660	343.92
13	344.65	0.549	0.706	0.908	0.711	344.9	0.892	0.697	345.32
14	344.13	0.593	0.754	0.929	0.689	344.01	0.916	0.672	344.40
15	345.18	0.518	0.660	0.891	0.727	345.53	0.872	0.716	345.95
16	345.95	0.480	0.602	0.868	0.747	346.24	0.845	0.740	346.67
17	346.28	0.468	0.580	0.860	0.754	346.47	0.836	0.748	346.89
18	346.63	0.451	0.565	0.847	0.764	346.77	0.822	0.759	347.18
19	346.95	0.429	0.529	0.831	0.776	347.12	0.804	0.773	347.52
20	347.25	0.409	0.500	0.815	0.787	347.43	0.786	0.786	347.81
21	347.55	0.389	0.458	0.798	0.799	347.72	0.767	0.800	348.07
22	347.77	0.362	0.435	0.772	0.815	348.07	0.739	0.818	348.37
23	348.10	0.334	0.375	0.744	0.832	348.35	0.710	0.836	348.59
24	348.35	0.299	0.329	0.705	0.853	348.61	0.669	0.860	348.77
25	348.56	0.263	0.289	0.660	0.875	348.73	0.625	0.884	348.80
26	348.65	0.232	0.239	0.618	0.895	348.72	0.584	0.904	348.70
27	348.62	0.199	0.197	0.569	0.915	348.56	0.537	0.924	348.46
28	348.28	0.155	0.134	0.499	0.941	348.11	0.473	0.950	347.94
29	347.88	0.123	0.101	0.443	0.959	347.61	0.424	0.966	347.43
30	347.57	0.107	0.084	0.414	0.967	347.32	0.399	0.974	347.14
31	347.23	0.090	0.065	0.383	0.975	346.99	0.374	0.980	346.82
32	346.87	0.074	0.047	0.352	0.983	346.64	0.348	0.986	346.49
33	346.67	0.061	0.039	0.328	0.987	346.35	0.329	0.990	346.22
34	346.46	0.050	0.032	0.307	0.991	346.09	0.312	0.993	345.99
35	346.23	0.040	0.025	0.288	0.994	345.86	0.297	0.996	345.78
36	346.03	0.032	0.019	0.273	0.996	345.67	0.284	0.997	345.61
37	345.60	0.018	0.010	0.246	0.999	345.35	0.263	0.999	345.32
38	344.95	0.000	0.000	0.212	1.000	344.95	0.236	1.000	344.95

 $u(T) = \pm 0.1$ K, and $u(x_i) = u(y_i) = \pm 0.001$.

$$Z_A = \frac{1 + (1 + 4k_i p_i^s)^{1/2}}{1 + [1 + 4k_i p_i^s (2 - y_i)]^{1/2}} \quad (4)$$

and Z_N is used for the non-associating component:**Fig. 2.** T - x - y diagram for TFA(1)–H₂O(2) at 101.3 kPa; ■, experimental result; —, calculated based on the NRTL model; ---, calculated based on the Wilson model.**Fig. 3.** T - x - y diagram for ETFA(1)–EtOH(2) at 101.3 kPa; ■, experimental result; —, calculated based on the NRTL model; ---, calculated based on the Wilson model.**Fig. 4.** T - x - y diagram for ETFA(1)–TFA(2) at 101.3 kPa; ■, experimental result; —, calculated based on the NRTL model; ---, calculated based on the Wilson model.**Table 5**

Antoine constants of the compounds.

Components	Antoine constants		
	A	B	C
TFA [24]	15.19548	3243.6286	−38.289
EtOH [24]	16.8969	3803.98	−41.68
ETFA [2]	23.9761	10081.33	185.03
H ₂ O [24]	16.2886	3816.44	−46.13

Antoine equation: $\ln(p_i^s/kPa) = A - B/(T/K + C)$ **Table 6**

Thermodynamic consistency test.

Systems	NRTL			Wilson		
	D	J	D−J	D	J	D−J
TFA–H ₂ O	5.47	14.81	−9.34	8.87	14.81	−5.94
ETFA–EtOH	1.89	9.52	−7.63	5.52	9.52	−4.00
ETFA–TFA	0.22	5.81	−5.59	1.46	5.81	−4.35

Table 7
Binary interaction parameters of the NRTL and Wilson model.

Model	Model parameter ^a			RMSD	
				y ^b	T(K) ^c
TFA (1) + H ₂ O (2)					
NRTL	$\Delta g_{12} = -660.5 \text{ J mol}^{-1}$	$\Delta g_{21} = -5564.1 \text{ J mol}^{-1}$	$\alpha = 0.33$	0.0176	0.719
Wilson	$\Lambda_{12} = -8658.1 \text{ J mol}^{-1}$	$\Lambda_{21} = 2445.5 \text{ J mol}^{-1}$	—	0.0264	0.998
ETFA (1) + EtOH (2)					
NRTL	$\Delta g_{12} = 4231.8 \text{ J mol}^{-1}$	$\Delta g_{21} = 2342.4 \text{ J mol}^{-1}$	$\alpha = 0.375$	0.0332	0.411
Wilson	$\Lambda_{12} = 648.1 \text{ J mol}^{-1}$	$\Lambda_{21} = 7297.2 \text{ J mol}^{-1}$	—	0.0352	0.487
ETFA (1) + TFA(2)					
NRTL	$\Delta g_{12} = -3777.8 \text{ J mol}^{-1}$	$\Delta g_{21} = 2879.1 \text{ J mol}^{-1}$	$\alpha = 0.47$	0.0216	0.473
Wilson	$\Lambda_{12} = 2782.8 \text{ J mol}^{-1}$	$\Lambda_{21} = -2750.8 \text{ J mol}^{-1}$	—	0.0210	0.264

^a Δg_{12} and Δg_{21} are the interaction parameters for NRTL, where $\Delta g_{12} = g_{12} - g_{11}$, $\Delta g_{21} = g_{21} - g_{22}$. Λ_{12} and Λ_{21} are the interaction parameters for Wilson.

^b $\text{RMSD} = (\sum_{i=1}^N (y_{\text{exp}} - y_{\text{cal}})^2 / N)^{1/2}$

^c $\text{RMSD} = (\sum_{i=1}^N (T_{\text{exp}} - T_{\text{cal}})^2 / N)^{1/2}$

$$z_N = \frac{2\{1 - y_i + [1 + 4kPy_i(2 - y_i)]^{1/2}\}}{(2 - y_i)\{1 + [1 + 4kPy_i(2 - y_i)]^{1/2}\}} \quad (5)$$

here k_i is the dimerization equilibrium constant for pure i , and k is the dimerization equilibrium constant for component i in the mixture, both of which are functions of temperature. Marek has pointed out that k_i is approximated with k . Therefore, the equilibrium constant of vapor-phase dimerization of TFA, K_2 , can be defined as the function of temperature by the expression obtained from the literature [27],

$$-\log k_2 = \frac{10.800 - 3053}{T} \quad (6)$$

3.3. Thermodynamic consistency test

Thermodynamic consistency test is to verify the reliability of the experimental data with the common thermodynamic equations. Semiempirical method recommended by Herington was adopted in this paper. Criteria of this method is that the difference between D and J must be less than 10. Here, D and J are calculated as follows:

$$D = 100 \times \frac{\left| \int_{x_1=0}^{x_1=1} \ln \gamma_1 / \gamma_2 dx_1 \right|}{\int_{x_1=0}^{x_2=1} \ln |\gamma_1 / \gamma_2| dx_1} \quad (7)$$

$$J = 150 \times \frac{T_{\max} - T_{\min}}{T_{\min}} \quad (8)$$

where $T_{\max}$ and $T_{\min}$ are the highest and lowest boiling point of the system, respectively. The test results of thermodynamic consistency for the three binary systems are listed in Table 6. From Table 6, it can be determined that all the experimental VLE data satisfied the thermodynamic consistency.

3.4. Correlation of vapor–liquid equilibria

In this work, the experimental VLE data were correlated by means of NRTL and Wilson models, respectively. The binary interaction parameters of both models were determined by the minimization of the objective function F using the simplex method as follows:

$$F = \sum_{i=1}^N \left(\frac{y_{i,\text{cal}} - y_{i,\text{exp}}}{y_{i,\text{exp}}} \right)^2 \quad (9)$$

where N is the number of experimental points, $y_{i,\text{cal}}$ and $y_{i,\text{exp}}$ are the calculated and experimental vapor phase mole fraction of component i , respectively. The correlated binary interaction parameters from experimental data are shown in Table 7.

From Figs. 2–4, it is found that there are intersections between the two equilibria curves (bubble-point line and dew-point line), which indicates that azeotropes exist in these three binary systems. Therefore, it is not easy to separate ETFA + TFA, ETFA + EtOH, and TFA + H₂O by distillation. In the esterification reactions of TFA with EtOH, it isn't economic to use excess TFA because TFA is an expensive material. But if EtOH is excess, it can't be completely separated from the ETFA because EtOH can form an azeotrope with ETFA and its azeotropic temperature is close to the boiling point of ETFA. In order to obtain desired ETFA, hence, adopting a combined process (such as distillation, extraction, decant, et al.) is required.

Table 7 shows that the root mean square deviations (RMSD) of the boiling temperature and vapor phase mole fraction. It is observed that deviations in boiling temperature and vapor phase composition of NRTL and Wilson models are small for the three binary systems, and average absolute deviations of both models satisfactorily correlate with the isobaric VLE data.

4. Conclusions

The VLE data for the three binary mixtures (ETFA + TFA, ETFA + EtOH, and TFA + H₂O) at 101.3 kPa were measured in this article. The reliability of the apparatus was confirmed by measured the VLE data of methanol + isopropanol binary mixture. The VLE data measured from this study were verified by Semiempirical method and the good thermodynamic consistencies were obtained. The VLE data were correlated by both NRTL and Wilson activity coefficient models and the corresponding binary interaction parameters were estimated.

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References

- [1] D.Q. Xu, K. Prasad, O. Repic, T.J. Blacklock, *Tetrahedron Lett.* 36 (1995) 7357–7360.
- [2] Z.X. Huang, H.M. Jiang, L. Li, et al., *Chem. Thermodyn.* 86 (2015) 75–79.
- [3] S.G. Venezia-Mestre, A.C. Venice, G.G. Mirano-Venezia, U.S. Pat., 4916256.
- [4] H.X. Wang, L. Li, C.S. Ye, *Fluid Phase Equilib.* 342 (2013) 42–46.
- [5] L. Xiao, Q.L. Wang, H.X. Wang, T. Qiu, *Fluid Phase Equilib.* 344 (2013) 79–83.
- [6] C.S. Ye, X.L. Dong, W.J. Zhu, et al., *Fluid Phase Equilib.* 367 (2014) 45–50.
- [7] T. Kh, R.U.B. Gazizov, A.N.P. elyalov, *Zhurnal Obshchei Khimii.* 52 (1982) 487–491.
- [8] I.S. Voloshanovskii, A.E. Derkach, E.A. Krasnova, et al. U.S. Pat., 1409622.
- [9] T.Kh Gazizov, R.U. Belyalov, A.N. Pudovik, *Zhurnal Obshchei Khimii.* 52 (1982) 2724–2728.
- [10] E.T. McBee, D.L. Christman, *J. Am. Chem. Soc.* 77 (1955) 755–756.
- [11] Merrell R. Fenske, Jennings H. Jones, U.S. Pat., 2858331.
- [12] K.W. Boeddeker, G. Bengtson, H. Pingel, *J. Membr. Sci.* 54 (1990) 1–12.
- [13] R.W. Merriman, *J. Am. Chem. Soc.* 103 (1913) 628–636.
- [14] H.S. Lai, Y.F. Lin, C.H. Tu, *J. Chem. Thermodyn.* 68 (2014) 13–19.
- [15] B. González, N. Calvar, A. Domínguez, J. Tojo, *J. Chem. Thermodyn.* 39 (2007) 322–334.
- [16] Abdullah S. Alkindi, Yahya M. Al-Wahaibi, Ann H. Muggeridge, *J. Chem. Eng. Data* 53 (2008) 2793–2796.
- [17] A. Kreglewski, *Bull. Acad. Pol. Sci. Ser. Sci. Chim.* 10 (1962) 629–633.
- [18] E.A. Kauck, A.R. Diesslin, *Ind. Eng. Chem.* 43 (1951) 2332–2334.
- [19] G.Q. Liu, L.X. Ma, J. Liu, *Handbook of Chemistry and Chemical Property Data*, Chemical Industry Press, Beijing, 2002.
- [20] S. Filippi, L. Madrigali, G. Polacco, et al., *Polym. Eng. Sci.* 46 (2006) 139–152.
- [21] L.M. Omota, I. Olga, F. Omota, O. Ciocirlan, *Rev. Roum. Chim.* 54 (2009) 63–73.
- [22] H.X. Wang, X.W. Bu, J.B. Yang, et al., *J. Chem. Thermodyn.* 74 (2014) 85–90.
- [23] T. Hiaki, A. Kawai, *Fluid Phase Equilib.* 158–160 (1999) 979–989.
- [24] Z.X. Chen, F.Y. Gu, W.M. Hu, *Chemical Thermodynamic*, Chemical Industry Press, Beijing, 2010.
- [25] J. George Hayden, P. John O'Connell, *Ind. Eng. Chem. Process Des. Dev.* 14 (3) (1975) 209–216.
- [26] J. Marek, *Collect. Czech. Chem. Commun.* 20 (1955) 1490–1520.
- [27] R.E. Lundin, F.E. Harris, L.K. Nash, *J. Am. Chem. Soc.* 74 (1952) 4654–4656.